

8/31/2026

Assignment-4

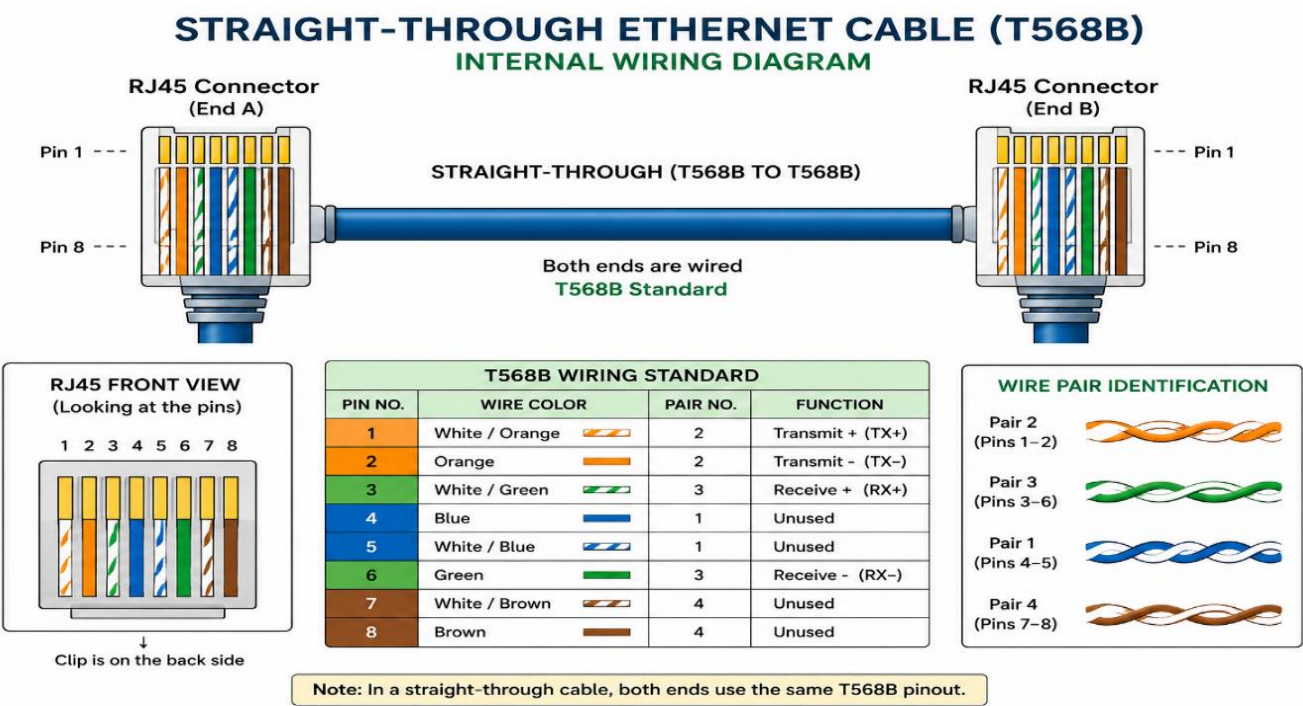
Created By
~pintu mer

1. List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

Point	Ethernet Cable	Fiber Optic Cable
1. Transmission	Uses electrical signals	Uses light signals
2. Speed & Distance	Good speed but shorter distance	Very high speed and much longer distance
3. Cost	Cheaper and easier to instal	More expensive and requires special equipment

Real-World Examples

- Ethernet Cable: Used to connect a home Wi-Fi router to a computer/PC.
 - Fiber Optic Cable: Used as a college campus backbone to connect different buildings at high speed.
2. Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.



3. Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

RJ45 Connector Crimping - Steps :

Required Tools:

- CAT6 cable, RJ45 connector, crimping tool, cable cutter, and LAN tester.

❖ Steps

1. Cut the CAT6 cable to the required length.
2. Remove the outer cover of the cable carefully.
3. Separate the 8 small wires inside the cable.
4. Arrange the wires in T568B order:
 - White-Orange
 - Orange
 - White-Green
 - Blue
 - White-Blue
 - Green
 - White-Brown
 - Brown
5. Straighten all 8 wires and keep them in the correct order.
6. Cut the wire ends equally so they are straight.
7. Put all 8 wires into the RJ45 connector.
8. Check the wire order one more time.
9. Put the RJ45 connector into the crimping tool.
10. Press the crimping tool firmly to fix the connector.
11. Check the connector to make sure the wires are properly fixed.
12. Test the cable using a LAN cable tester.

✓ **Result:** The RJ45 connector is successfully attached to the CAT6 cable and is ready for network use.

4. Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences. Hint: Consider speed, distance, and budget for your answer.

Gaming Café Setup

1. Cable:

- Use CAT6 Ethernet cable.

2. Tool:

- Use an RJ45 crimper.

3. Why:

- CAT6 gives good speed for LAN gaming and works well for short distances.

4. Budget:

- Ethernet is cheaper and easier to install than fiber optic.

5. Final Choice:

- CAT6 Ethernet + RJ45 crimper is best for 10 PCs.